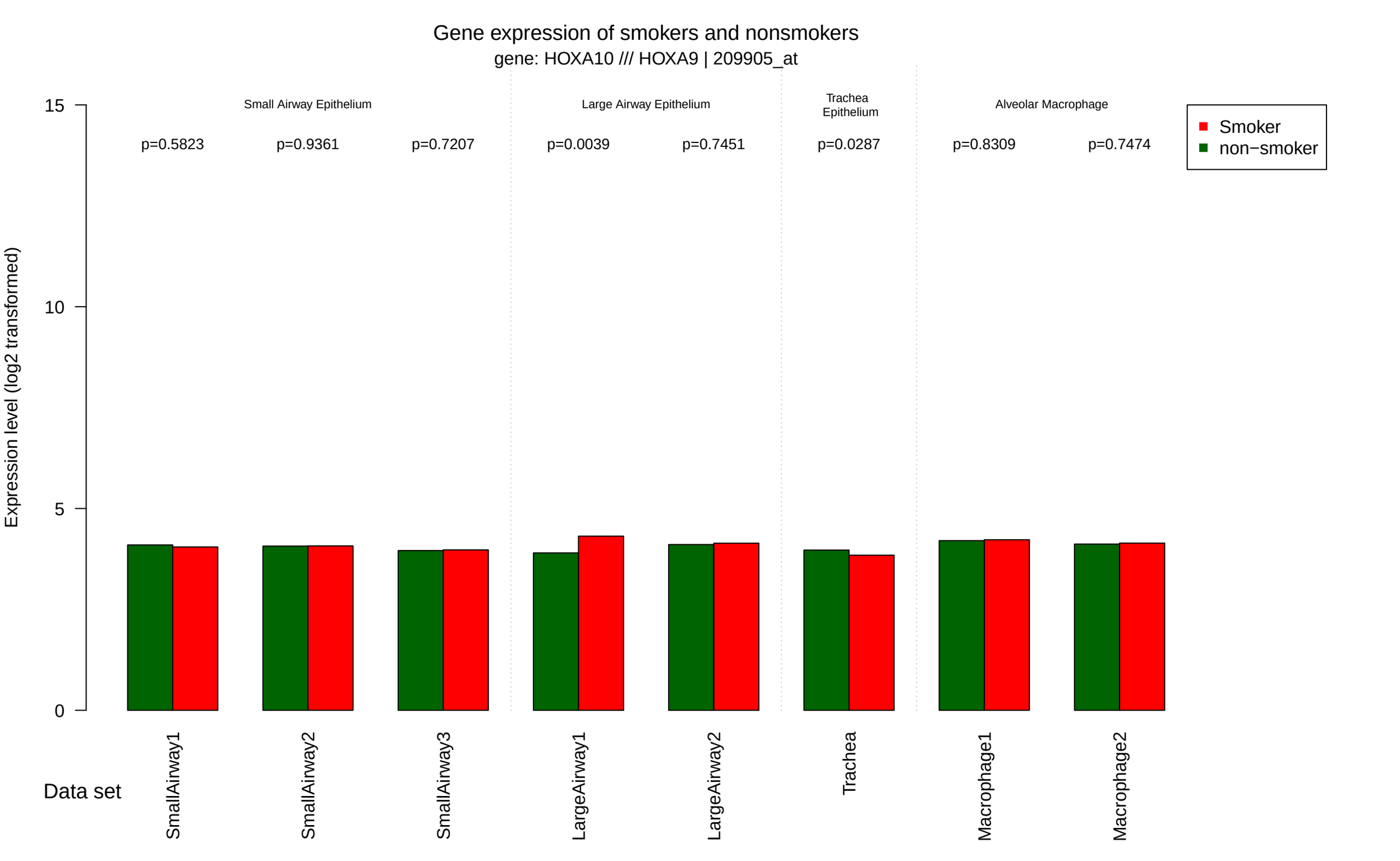
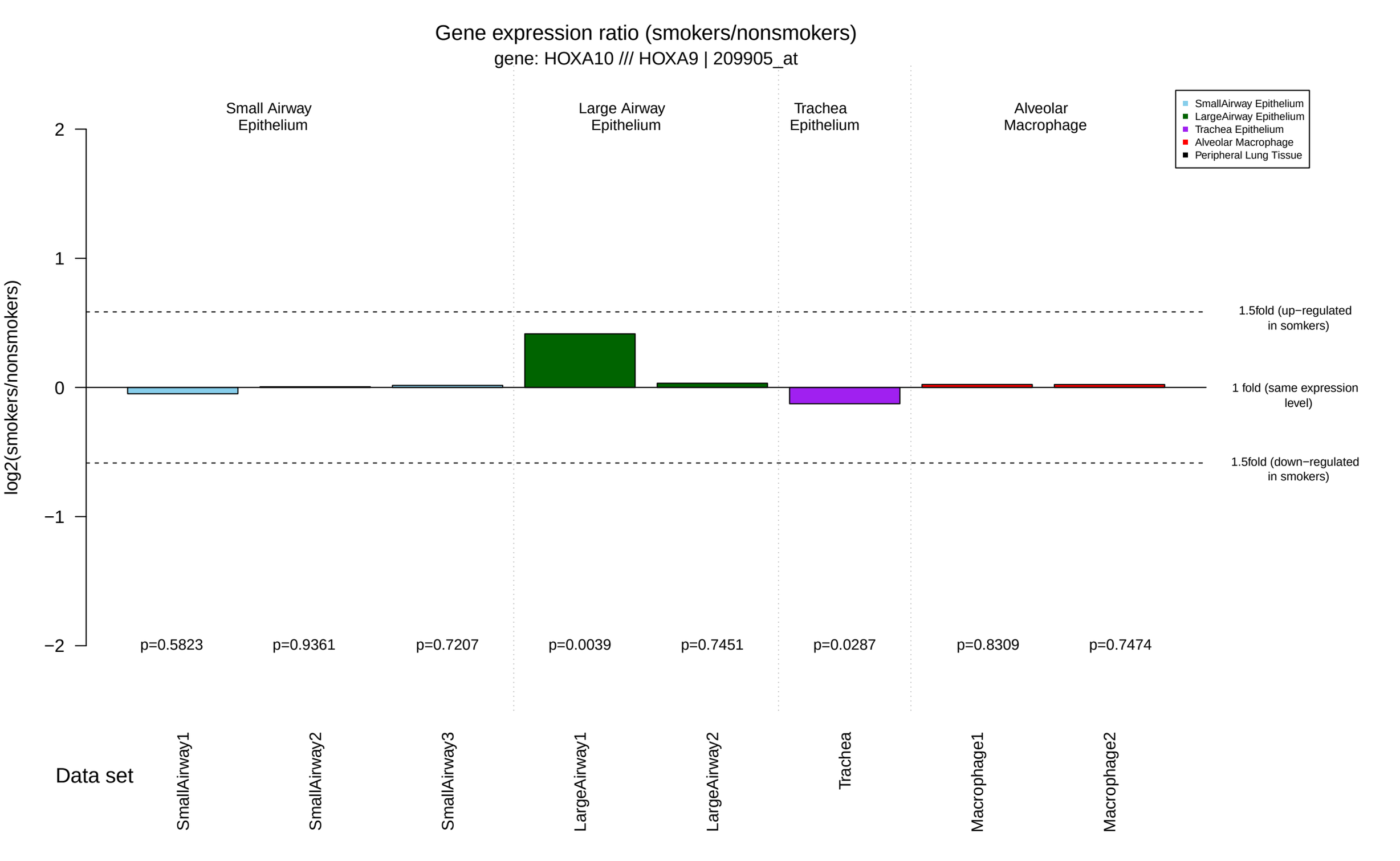
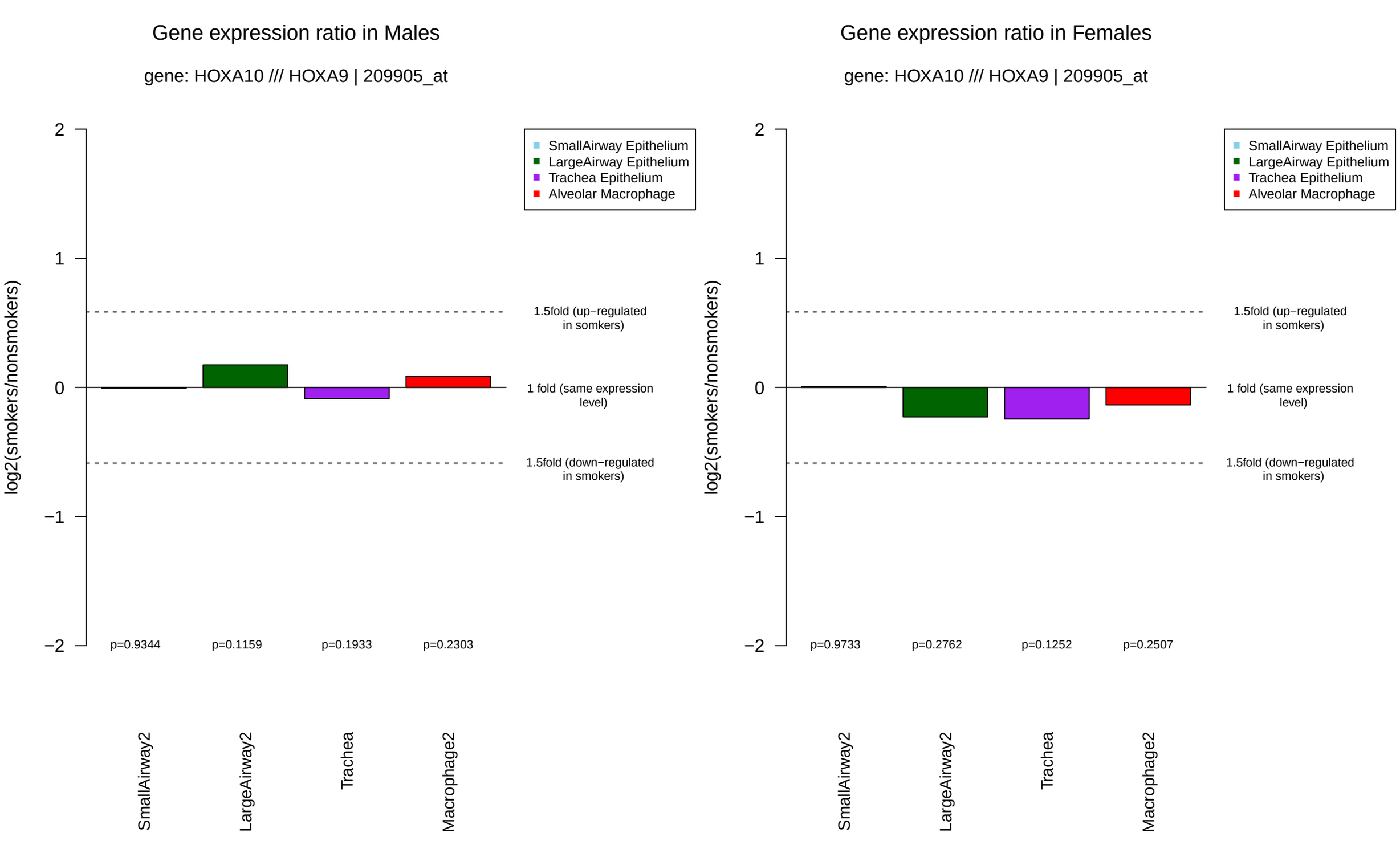
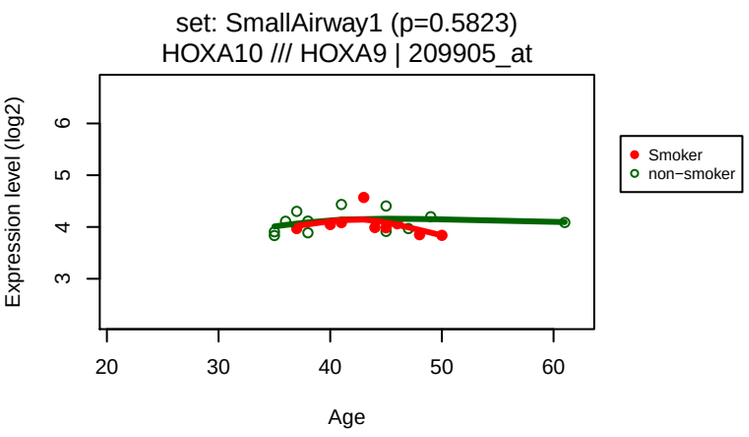
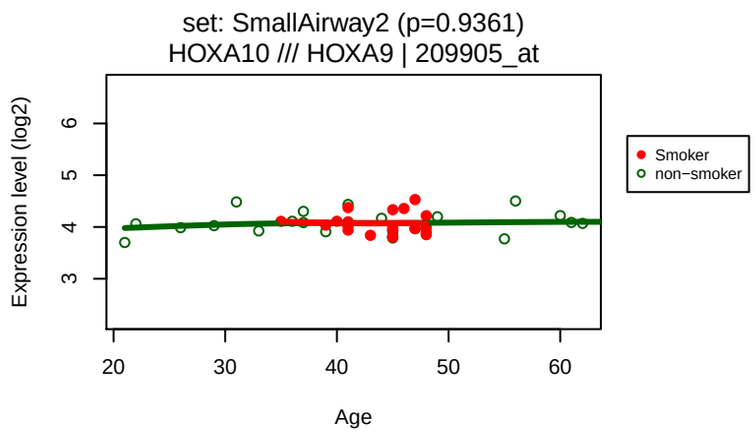
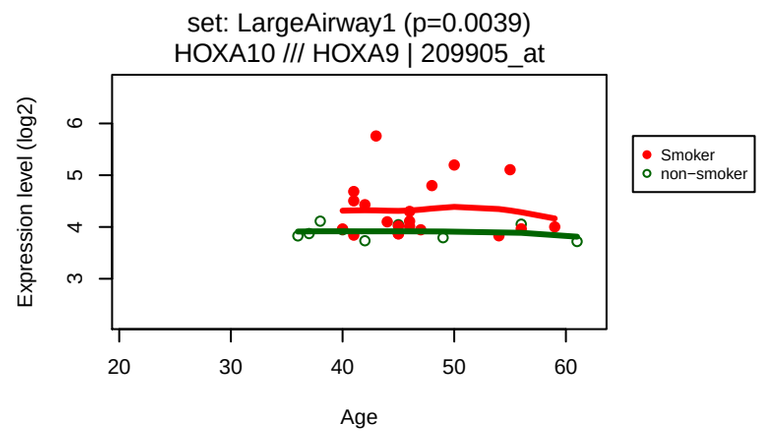

Gene expression of smokers and nonsmokers
gene: HOXA10 /// HOXA9 | 209905_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5823
p=0.9361
p=0.7207
p=0.0039
p=0.7451
p=0.0287
p=0.8309
p=0.7474
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

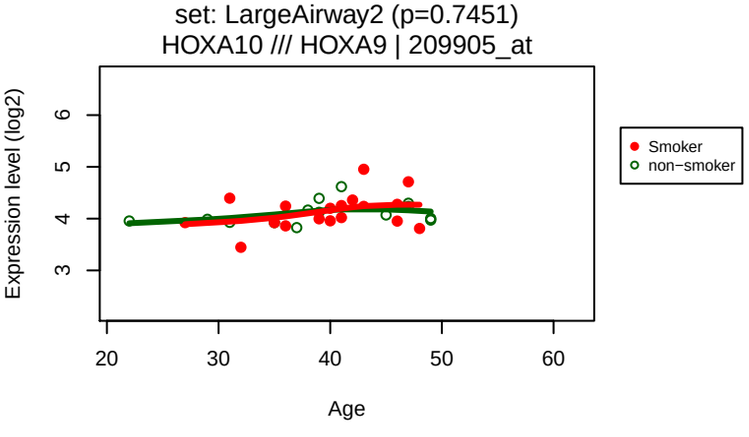

Gene expression ratio (smokers/nonsmokers)
gene: HOXA10 /// HOXA9 | 209905_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5823
p=0.9361
p=0.7207
p=0.0039
p=0.7451
p=0.0287
p=0.8309
p=0.7474
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

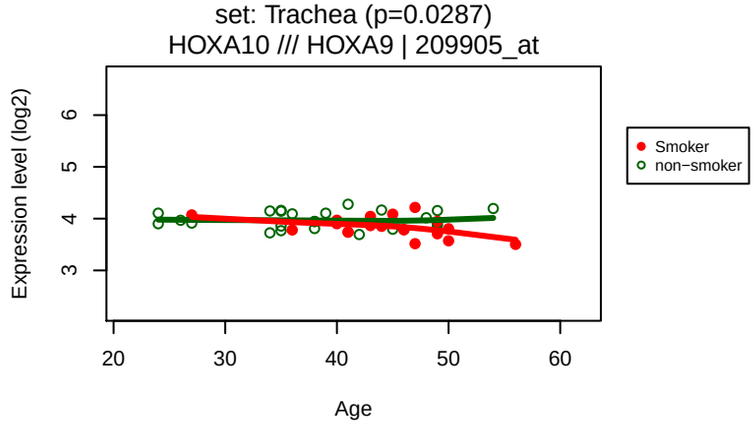

Gene expression ratio in Males
gene: HOXA10 /// HOXA9 | 209905_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9344
p=0.1159
p=0.1933
p=0.2303
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HOXA10 /// HOXA9 | 209905_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9733
p=0.2762
p=0.1252
p=0.2507
SmallAirway2
LargeAirway2
Trachea
Macrophage2

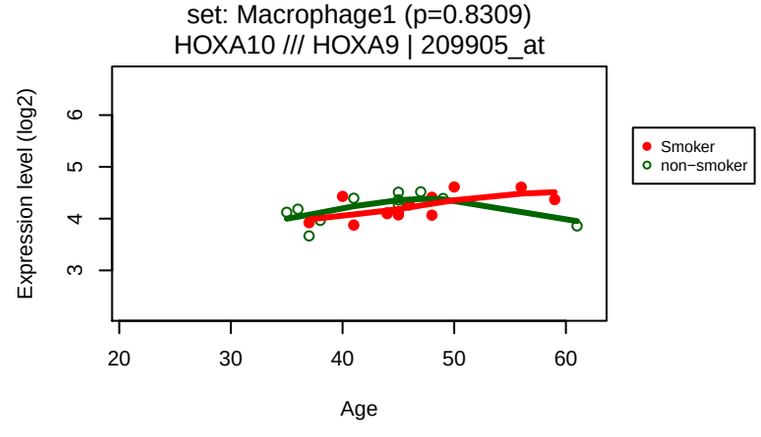

set: SmallAirway1 (p=0.5823)
HOXA10 /// HOXA9 | 209905_at
Expression level (log2)
Age
Smoker
non−smoker

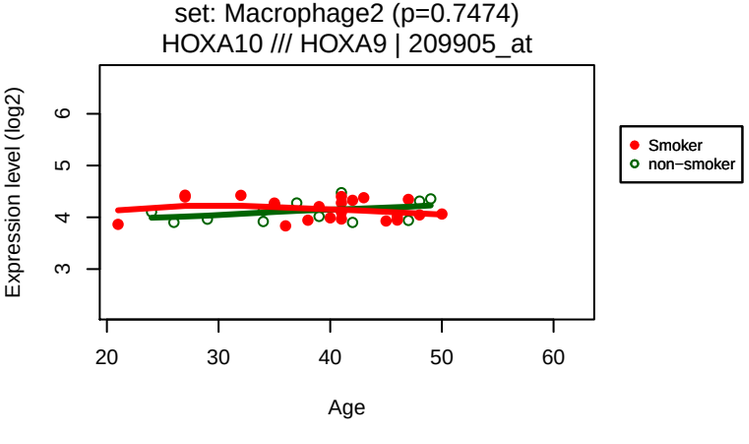

set: SmallAirway2 (p=0.9361)
HOXA10 /// HOXA9 | 209905_at
Expression level (log2)
Age
Smoker
non−smoker

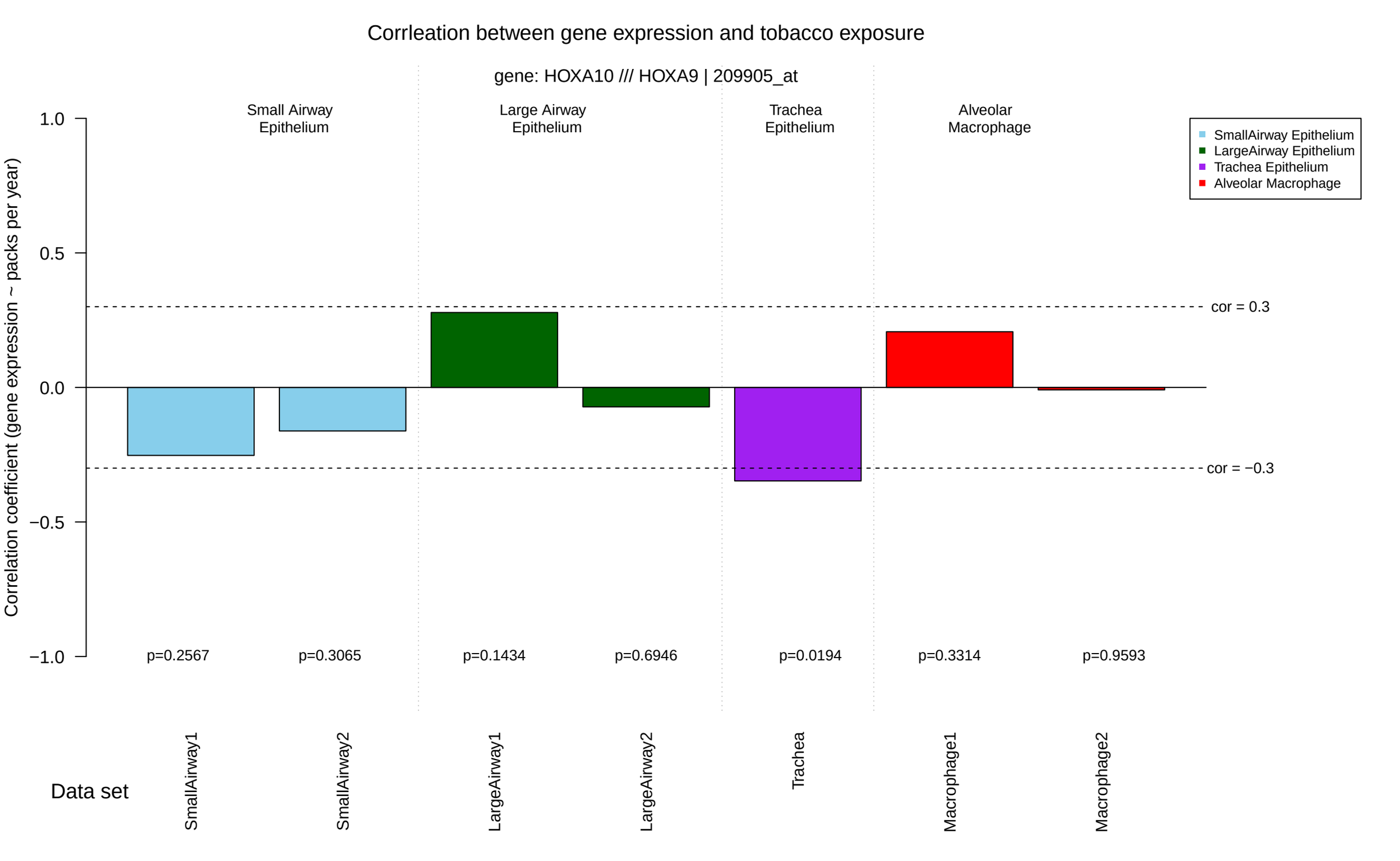

set: LargeAirway1 (p=0.0039)
HOXA10 /// HOXA9 | 209905_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7451)
HOXA10 /// HOXA9 | 209905_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0287)
HOXA10 /// HOXA9 | 209905_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8309)
HOXA10 /// HOXA9 | 209905_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7474)
HOXA10 /// HOXA9 | 209905_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HOXA10 /// HOXA9 | 209905_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.2567
p=0.3065
p=0.1434
p=0.6946
p=0.0194
p=0.3314
p=0.9593
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2